

PROBLEM SET 2

ECO403/503–INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

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Instructions: Due on 03/09/2021; scan or type your work and submit through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

1. Consider the insurance problem studied in the lecture “Lotteries with Monetary Prizes & Applications”.
 - (a) Show that if insurance is not actuarially fair, so that $q > \pi$, then the individual will not insure completely, i.e., $\alpha^* < D$.
 - (b) *Comparative Statics.* Again, suppose that insurance is not actuarially fair and assume that the agent chooses a positive level of insurance, i.e., $\alpha^* > 0$. How does α^* depends on the wealth level w and on the price of insurance q ? Use the implicit function theorem to answer the previous questions.
 - (c) Suppose now that you have two individuals whose preferences can be represented by the strictly increasing and strictly concave utility functions u_1 and u_2 . Suppose that individual 1 is more risk averse than individual 2. Show that if insurance is not actuarially fair, then individual 1 will buy more insurance than individual 2, i.e., $\alpha_1^* > \alpha_2^*$. You can assume interior solutions for the decision problems of individuals 1 and 2.
2. Suppose that an individual has a Bernoulli utility function $u(x) = \sqrt{x}$.
 - (a) Calculate the coefficient of absolute risk aversion at the level $x = 5$.
 - (b) Calculate the certainty equivalent and the risk premium for the lottery $(1/2)16 \oplus (1/2)4$.
 - (c) Calculate the certainty equivalent and the risk premium for the lottery $(1/2)36 \oplus (1/2)16$.
3. Consider a lottery of monetary outcomes $(1/2)[x + \epsilon] \oplus (1/2)[x - \epsilon]$, that is, this lottery the amount $x + \epsilon$ with probability $1/2$ and the amount $x - \epsilon$ with probability $1/2$.
 - (a) Compute the certainty equivalent $c(x, u)$ of this lottery, for any utility function u and any amount of money x .
 - (b) Compute the second derivative of the $c(x, u)$ with respect to ϵ . Show that the limit of this derivative as $\epsilon \rightarrow 0$ is exactly $-r_A(x, u)$.
4. Consider the asset allocation problem in lecture 4. Suppose that the utility function is given by

$$u(x) = -\exp(-\gamma x)$$

where $\gamma > 0$ is a parameter. Also, assume that the risky asset z is distributed according to a *normal distribution* with mean $\mu > 1$ and variance σ^2 . That is, the pdf of the random return on the asset z is given by

$$\phi(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2}\left(\frac{z - \mu}{\sigma}\right)^2\right]$$

- (a) Find explicitly the optimal portfolio (α, β) of this individual, i.e. find the demands α and β for the risky and safe asset respectively as a function of the parameters of the problem.
- (b) Compute the coefficient of absolute risk aversion for the utility function u in this problem. What is the meaning of γ ?
- (c) Suppose that you have two individuals, with the utility function described in this problem, but who differ on the parameter γ , say, individual 1 has parameter γ_1 and individual 2 has parameter γ_2 with $\gamma_1 > \gamma_2$. Using the previous item, what is the implication for each individual's demand for the risky asset? Verify your conclusion explicitly by computing the optimal demand for the risky asset for each individual.

5. Finish the proof of proposition 4.12 in the lecture notes by showing that points (2), (3) and (4) are equivalent. *Hint: Try showing the following chain: (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (2).*
6. Prove proposition 4.18. *Hint: The same techniques as in the proof of 4.14 can be used in here. You will need to use integration by parts twice.*
7. A **partial order** is a binary relation $\triangleright$ over a set X , such that $\triangleright$ is *reflexive, antisymmetric and transitive*. Prove that the binary relation $\triangleright_{\text{fosd}}$ over $\mathfrak{F}$ is a partial order.